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Towards an AI policy framework in scholarly publishing

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Abstract

The rapid adoption of AI tools in academic research raises pressing ethical concerns. I examine major publishing policies in science and medicine, uncovering inconsistencies and limitations in guiding AI usage. To encourage responsible AI integration while upholding transparency, I propose an enabling framework with author and reviewer policy templates.

Keywords: generative artificial intelligence (AI); large language models (LLMs); science; policy; publishing

Imperative of coherent AI policy

The rapid emergence of advanced artificial intelligence (AI) tools, especially large language models (LLMs), has fundamentally altered the scientific landscape, aiding in tasks like writing, data analysis, and even idea generation [1, 2]. However, as an author, reviewer, and editor, I have also witnessed firsthand the ethical complications introduced by these tools, including issues around originality, plagiarism, attribution, transparency, and reproducibility [3, 4]. To promote responsible and productive integration of AI in academic endeavors, the scientific community must act quickly and collectively to develop well-defined, realistic, and coherent policies [5].

In my experience both writing manuscripts with AI assistance [6] and reviewing submissions that disclose varying degrees of AI contribution, I have struggled to find consistent guidelines. Policies differ widely across publishers and organizations, and even within the same journal, policies may change rapidly (e.g., *Science* changed its AI policy on 16 November 2023, less than 10 months after its initial policy¹). The uncertainty and discrepancies risk undermining consistency and reproducibility, leaving authors confused about best practices.

Drawing from my experiences, I analyze AI policies from leading scholarly organizations, identifying shared elements, critical inconsistencies, and major oversights. Building on the strengths of existing policies and addressing their limitations, I propose a pragmatic, enabling policy framework—with example author and reviewer policy templates—to provide a foundation for building consensus.

Policy commonalities, inconsistencies, and limitations

Scientific organizations in disciplines spanning psychology, management, and neuroscience currently lack AI guidelines [4]. I focus on three major associations dedicated to journal publishing guidelines—namely, the International Committee of Medical Journal Editors

(ICMJE)ⁱⁱ, Committee on Publication Ethics (COPE)ⁱⁱⁱ, and World Association of Medical Editors (WAME)^{iv}—as well as five leading medical and science journal groups: the Journal of the American Medical Association and its network of journals (“JAMA”)^v, the Lancet group (“Lancet”)^{vi}, Nature and Springer Nature journals (“Nature”)^{vii}, Science family of journals (“Science”)ⁱ, and Proceedings of the National Academy of Sciences journals (“PNAS”)^{viii}. They were selected based on their prominence and global influence on scientific publishing. For example, the ICMJE policy was adopted by journals such as *NEJM (New England Journal of Medicine)*^{ix}; the WAME policy was adapted by journals such as *BMJ (British Medical Journal)*^x; and the *Lancet* policy was similar to that of Elsevier (including Cell Press)^{xi}. I categorize policies for authors and reviewers, organizing them along several critical dimensions, including AI authorship, the scope of permitted AI use, and reporting requirements.

Author policy

As Table 1 shows, there is consensus that AI systems cannot receive authorship credit, placing the onus of quality control squarely on human researchers. But guidance on upholding this responsibility varies. For example, ICMJE, WAME, and *Science* outline specific actions for authors, such as reviewing and editing for accuracy, plagiarism, and proper attribution. On the other hand, *PNAS* and *Lancet* emphasize the importance of fact-checking. Other policies lack detailed instructions.

A more fundamental difference, however, lies in the scope of AI usage. While five organizations impose no explicit restrictions, three do. *Nature* prohibits the use of AI-generated images and videos, citing unresolved “legal issues.” *Lancet* goes a step further, stipulating that AI can only be used to “improve readability and language of the work,” not replace “researcher tasks.” Originally, *Science* had the most restrictive policy, forbidding “[t]ext generated from AI” along with “figures, images, [and] graphics.” However, this policy was amended on 16 November 2023, making it more similar to *Nature*’s. The limitation to “readability and language” by *Lancet* (and Elsevier) creates inconsistency, which potentially may also be discriminatory against researchers who find advanced editing services inaccessible. If authors can employ human editors for more in-depth refinement than “readability and language,” there seems to be no fundamental reason to prohibit the same use of AI.

Another area of policy divergence concerns guidance on disclosing AI usage. Disclosure involves three key elements. The first is the identification of the AI tool used (“what to include”). *JAMA* provides details instructions, but others offer limited guidance. The second element is specifying the AI tool’s contribution to content creation (“how to report”). WAME covers the extent and type of use, prompts, and date/time. *JAMA* instructs authors to describe the generated content that was included. *Science* asks for the prompts used. The five other organizations issue only a generic request to describe AI use. And none of them address the issue of citation—such as when authors quote or paraphrase text generated by AI. The third element pertains to reporting locations (“where to report”), which run from cover letters, methods sections, abstracts, and acknowledgments to statements at the end of the article. This variation can be confusing for authors and hinder the gathering of AI usage data across publications.

A critical issue missing from all the policies is whether disclosure requirements should differ based on the level of AI involvement. A reasonable argument can be made that detailed reporting should only be mandated when AI systems are used in capacities that exceed typical human editing services, just as authors are generally not required to disclose their use of university writing centers or professional editors. This approach strikes a balance between transparency and reporting burdens—a key consideration for widespread acceptance.

The policies also largely overlook reproducibility—the ability to replicate AI-assisted tasks for consistent results [4]. Although specifying the AI tool and prompts used is important, the stochastic nature of AI algorithms presents unique challenges. The same tool and prompt can produce different outcomes. Moreover, because AI involvement in research can be extensive and iterative, and because AI algorithms may be updated during the project, complete documentation may be impractical. This multitude of challenges renders complete reproducibility cumbersome if not impossible. Perhaps a more pragmatic approach is to focus on documenting essential details without overwhelming researchers.

Reviewer policy

For reviewers, confidentiality considerations drive *JAMA*, *PNAS*, *Lancet*^{xii}, and *Science*^{xiii} to prohibit the submission of any part of the manuscript or related materials, such as reviewer comments, to AI software—even for the purpose of readability and language improvement. Specifically, *Science* disallows any AI usage, while WAME does not mention reviewer comments. ICMJE, COPE, and *Nature* offer no explicit guidelines for reviewers. Most organizations are vague about disclosure requirements, except *JAMA*, which has a reporting policy similar to its guidelines for authors.

Nonetheless, many reviewers might not be well-versed in the nuances of data privacy. For instance, interactions with ChatGPT become part of its training data unless users opt out via “Chat history & training” or use ChatGPT Enterprise. Therefore, to help reviewers better understand the implications of using AI tools, journals should consider providing explicit guidelines to explain these limitations before the review process starts.

Towards a pragmatic, enabling policy framework

The analysis shows that current policy development is fraught with inconsistencies and ambiguities, complicating researchers’ efforts to maintain ethical consistency across publishing venues. The absence of AI policy in many journals creates a void, leaving the scholarly community vulnerable to ethical lapses and inconsistencies in ethical standards. Therefore, building an informed consensus within the academic community is becoming an urgent priority—crucial for establishing norms and promoting best practices.

The items outlined in **Table 1** provide a structure for doing so. I argue that circumscribing the role of AI—such as confining AI to merely enhancing the “readability and language” of the work, or to language tasks for that matter—is neither pragmatic nor constructive. It is not pragmatic because it is unenforceable^{xiv}. As a result, stringent restrictions might even discourage researchers from fully disclosing their use of AI, thereby undermining the transparency of the research literature. It is also not constructive because it overlooks many nuances and negates the real value provided by AI. These constraints hamper its capacity to, for example, provide feedback and substantive editing, as well as to augment analytical work [1].

Instead, I propose an *enabling* principle: AI policy should aim to encourage the exploitation of AI capacities to help researchers produce their best work, fostering transparency and responsible integration in the process. Given the challenges in detecting AI-created content and the impracticality of enforcing restrictive policies—which could even backfire—I further propose a *pragmatic* principle. That is, policy should focus not on constraining AI usage, but on proper disclosure when AI is used beyond typical human editing services. Only through pragmatic, enabling policies can we truly harness the transformative power of AI while upholding ethics.

Striking a balance between transparency and ease of implementation, I offer an initial policy that implements this pragmatic, enabling approach and combines the strengths of ICMJE,

WAME, and *JAMA* guidelines in **Box 1**. Although the challenge of reproducibility remains, using shared chat history, such as that provided by ChatGPT, may capture enough details without overburdening authors.

Beyond policy formulation, effective implementation demands dedicated resources and ongoing efforts [7]. Through continued collaboration and collective oversight, we can harness the potential of AI to accelerate discovery fruitfully and ethically. The approach and policy framework introduced here lay the groundwork for this exciting new partnership between human ingenuity and machine intelligence.

Acknowledgments

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Resources

- ⁱ<https://www.science.org/content/blog-post/change-policy-use-generative-ai-and-large-language-models>
- ⁱⁱ<https://www.icmje.org/recommendations/browse/roles-and-responsibilities/defining-the-role-of-authors-and-contributors.html#four>
- ⁱⁱⁱ<https://publicationethics.org/cope-position-statements/ai-author>
- ^{iv}<https://wame.org/page3.php?id=106>
- ^v<https://jamanetwork.com/journals/jama/fullarticle/2807956>
- ^{vi}<https://www.thelancet.com/preparing-your-manuscript>
- ^{vii}<https://www.nature.com/nature-portfolio/editorial-policies/ai>
- ^{viii}<https://www.pnas.org/post/update/pnas-policy-for-chatgpt-generative-ai>
- ^{ix}<https://www.nejm.org/about-nejm/editorial-policies>
- ^x<https://authors.bmj.com/policies/ai-use/>
- ^{xi}<https://www.elsevier.com/about/policies-and-standards/publishing-ethics>
- ^{xii}<https://www.thelancet.com/peer-review>
- ^{xiii}<https://www.science.org/content/page/science-information-reviewers>
- ^{xiv}<https://help.openai.com/en/articles/8313351-how-can-educators-respond-to-students-presenting-ai-generated-content-as-their-own>

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Table 1: Summary of current AI policies for authors and reviewers as of 11 December 2023^a

Item	ICMJE	COPE	WAME	JAMA	PNAS	Science	Nature	Lancet
Time of policy in 2023	May	Feb.	May (updated from Jan.)	Jul. (updated from Jan.)	Feb.	Nov. (updated from Jan.)	Jan.	Jun.
Authors								
Scope of AI use	Any	Any	Any	Any	Any	No generate images without editor permission	No generative images	Readability and language only
Authorship & responsibility	Humans	Humans	Humans	Humans	Humans	Humans	Humans	Humans
Suggested actions	Review, edit; ensure no plagiarism, appropriate attribution	-	Ensure accuracy, balanced view, no plagiarism, appropriate attribution	-	Fact-check	Review; guard against bias; ensure no plagiarism, appropriate citations,	-	Review, edit
What to include	-	Tool name	Tool name, version	Tool name, version and extension, manufacturer	-	Tool name, version	-	-
How to report	Describe use	Disclose use	Indicate extent and type of use, prompts, date/time	Describe generated content that was included	Note use	Full prompts	Document use	Disclose use
Where to report	Cover letter & manuscript	Methods	Acknowledgment (for text); Abstract & Methods (for analysis, code, results)	Acknowledgment or Methods	Methods (if unavailable, Acknowledgment)	Cover letter, Acknowledges & Methods	Methods (if unavailable, a suitable part)	Statement at the end of the article
Reviewers								
Scope of AI use	-	-	No manuscript	No manuscript or review	No manuscript or review	Not allowed	-	No manuscript or review
What to include	-	-	-	Tool name, version and extension, manufacturer	-	-	-	-
How to report	-	-	Explain use	Describe the generated content	-	-	-	-

^aRelevant policy content from each organization is available online as supplemental materials (<https://osf.io/ra9e8/>). ICMJE, International Committee of Medical Journal Editors; COPE, Committee on Publication Ethics; WAME, World Association of Medical Editors; JAMA, Journal of the American Medical Association; PNAS, Proceedings of the National Academy of Sciences.

Box 1. Developing AI policy templates for authors and reviewers

As AI tools become increasingly integrated into the research and publishing workflow, the scientific community would benefit from developing clear, consistent AI policies and norms. Below is an illustration of potential policy elements to consider when forming AI guidelines.

For authors

Clearly delineating authorship and responsibility can help maintain transparency. For instance, the policy could state that AI systems do not qualify for authorship, and authors should take full accountability for their submissions, including the appropriate scrutiny and attribution of any AI-generated content.

Reporting guidelines may cover the disclosure of AI use beyond typical writing assistance services, asking authors to specify the AI tool, version, purpose, and extent of usage. Language could suggest that authors 1) indicate sections that enlist AI assistance, 2) outline the goal and incorporation process for generated text, and 3) provide original prompts given to AI systems.

Potential locations for such AI disclosures include the Acknowledgements, Methods, or other relevant sections, depending on the type of AI contribution.

For reviewers

Sample policy language could reaffirm reviewer responsibilities of confidentiality, noting that manuscript information should not be entered into AI systems without consent. When AI tools are used without violating confidentiality, policy may ask reviewers to report the name, version, developer, and purpose of usage.

The examples and considerations above aim to spur thoughtful discussion around developing ethical, constructive AI policies across scientific publications and organizations. They are not intended to be prescriptive, but to illustrate reporting details and norms that may help guide transparency. To help implement these policy elements, I have also provided policy templates that can be adopted or adapted, available online as supplemental materials (<https://osf.io/ra9e8/>).

Supplementary Materials for
Towards an AI policy framework in scholarly publishing

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The PDF file includes:

Pages 2 to 8	Materials
Page 9	Templates of AI policies for authors and reviewers

Materials

1 ICMJE (International Committee of Medical Journal Editors), May 2023

1.1 For authors

<https://www.icmje.org/recommendations/browse/roles-and-responsibilities/defining-the-role-of-authors-and-contributors.html#four>

At submission, the journal should require authors to disclose whether they used artificial intelligence (AI)-assisted technologies (such as Large Language Models [LLMs], chatbots, or image creators) in the production of submitted work. Authors who use such technology should describe, in both the cover letter and the submitted work, how they used it. Chatbots (such as ChatGPT) should not be listed as authors because they cannot be responsible for the accuracy, integrity, and originality of the work, and these responsibilities are required for authorship (see Section II.A.1). Therefore, humans are responsible for any submitted material that included the use of AI-assisted technologies. Authors should carefully review and edit the result because AI can generate authoritative-sounding output that can be incorrect, incomplete, or biased. Authors should not list AI and AI-assisted technologies as an author or co-author, nor cite AI as an author. Authors should be able to assert that there is no plagiarism in their paper, including in text and images produced by the AI. Humans must ensure there is appropriate attribution of all quoted material, including full citations.

1.2 For reviewers

Not available.

2. COPE (Committee on Publication Ethics), 13 February 2023

2.1 For authors

<https://publicationethics.org/cope-position-statements/ai-author>

AI tools cannot meet the requirements for authorship as they cannot take responsibility for the submitted work. As non-legal entities, they cannot assert the presence or absence of conflicts of interest nor manage copyright and license agreements.

Authors who use AI tools in the writing of a manuscript, production of images or graphical elements of the paper, or in the collection and analysis of data, must be transparent in disclosing in the Materials and Methods (or similar section) of the paper how the AI tool was used and which tool was used. Authors are fully responsible for the content of their manuscript, even those parts produced by an AI tool, and are thus liable for any breach of publication ethics.

2.2 For reviewers

Not available.

3. WAME (World Association of Medical Editors), 31 May 2023 (updated from 20 January 2023)

3.1 For authors

<https://wame.org/page3.php?id=106>

WAME Recommendation 1: *Chatbots cannot be authors.* Journals have begun to publish articles in which chatbots such as Bard, Bing and ChatGPT have been used, with some journals listing chatbots as co-authors. The legal status of an author differs from country to country but under most jurisdictions, an author must be a legal person. Chatbots do not meet the International

Committee of Medical Journal Editors (ICMJE) authorship criteria, particularly that of being able to give “final approval of the version to be published” and “to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.” (10) No AI tool can “understand” a conflict-of-interest statement, and does not have the legal standing to sign a statement. Chatbots have no affiliation independent of their developers. Since authors submitting a manuscript must ensure that all those named as authors meet the authorship criteria, chatbots cannot be included as authors.

WAME Recommendation 2: *Authors should be transparent when chatbots are used and provide information about how they were used.* The extent and type of use of chatbots in journal publications should be indicated. This is consistent with the ICMJE recommendation of acknowledging writing assistance (11) and providing in the Methods detailed information about how the study was conducted and the results generated. (12)

WAME Recommendations 2.1: *Authors submitting a paper in which a chatbot/AI was used to draft new text should note such use in the acknowledgment; all prompts used to generate new text, or to convert text or text prompts into tables or illustrations, should be specified.*

WAME Recommendation 2.2: *When an AI tool such as a chatbot is used to carry out or generate analytical work, help report results (e.g., generating tables or figures), or write computer codes, this should be stated in the body of the paper, in both the Abstract and the Methods section. In the interests of enabling scientific scrutiny, including replication and identifying falsification, the full prompt used to generate the research results, the time and date of query, and the AI tool used and its version, should be provided.*

WAME Recommendation 3: *Authors are responsible for material provided by a chatbot in their paper (including the accuracy of what is presented and the absence of plagiarism) and for appropriate attribution of all sources (including original sources for material generated by the chatbot).* Authors of articles written with the help of a chatbot are responsible for the material generated by the chatbot, including its accuracy. Noting that plagiarism is “the practice of taking someone else's work or ideas and passing them off as one's own” (13), not just the verbatim repetition of previously published text. It is the author's responsibility to ensure that the content reflects the author's data and ideas and is not plagiarism, fabrication or falsification. Otherwise, it is potentially scientific misconduct to offer such material for publication, irrespective of how it was written. Similarly, authors must ensure that all quoted material is appropriately attributed, including full citations, and that the cited sources support the chatbot's statements. Since a chatbot may be designed to omit sources that oppose viewpoints expressed in its output, it is the authors' responsibility to find, review and include such counterviews in their articles. (Of course, such biases are also found in human authors.) Authors should identify the chatbot used and the specific prompt (query statement) used with the chatbot. They should specify what they have done to mitigate the risk of plagiarism, provide a balanced view, and ensure the accuracy of all their references.

WAME Recommendation 4: *Editors and peer reviewers should specify, to authors and each other, any use of chatbots in the evaluation of the manuscript and generation of reviews and correspondence. If they use chatbots in their communications with authors and each other, they should explain how they were used.* Editors and reviewers are responsible for any content

and citations generated by a chatbot. They should be aware that chatbots retain the prompts fed to them, including manuscript content, and supplying an author's manuscript to a chatbot breaches confidentiality of the submitted manuscript.

WAME Recommendation 5: *Editors need appropriate tools to help them detect content generated or altered by AI. Such tools should be made available to editors regardless of ability to pay for them, for the good of science and the public, and to help ensure the integrity of healthcare information and reducing the risk of adverse health outcomes.* Many medical journal editors use manuscript evaluation approaches that were not designed to deal with AI innovations and industries, including manipulated plagiarized text and images and papermill-generated documents. They have already been at a disadvantage when trying to differentiate the legitimate from the fabricated, and chatbots take this challenge to a new level. Editors need access to tools that will help them evaluate content efficiently and accurately. This is of particular importance to editors of medical journals where the adverse consequences of misinformation include potential harms to people.

3.2 For reviewers

Recommendation 4 above.

4. JAMA and its network journals, 27 July 2023 (updated from 31 January 2023)

4.1 For authors

<https://jamanetwork.com/journals/jama/fullarticle/2801170>

<https://jamanetwork.com/journals/jama/fullarticle/2807956>

Did you use AI, a language model, machine learning, or similar technologies to create or assist with creation or editing of any of the content in this submission (eg, text, tables, figures, video)? (Note: this does not include basic tools for checking grammar, spelling, references, etc.)

Please provide a description of the AI-generated content that is included in this submission and the name of the model or tool used, version and extension numbers, and manufacturer in the space below.

Please confirm that you take responsibility for the integrity of the content generated by these tools and that you have provided a description of such generated content and the name of the model or tool used, version and extension numbers, and manufacturer in the Acknowledgment or Methods section of the manuscript.

4.2 For reviewers

<https://jamanetwork.com/journals/jama/fullarticle/2807956>

Entering any part of the manuscript or abstract or the text of your review into a chatbot, language model, or similar tool is a violation of our confidentiality agreement.

If you used an AI tool as a resource for your review in a way that does not violate the journal's confidentiality policy, you must provide the name of the tool and how it was used.

Please remember that you are ultimately responsible for all of the content of this review.

Did you use a chatbot, language model, or similar tool as a resource during your review? As a reminder, our confidentiality policy prohibits the entering of any part of the manuscript or your review into a chatbot, language model, or similar tool.

If yes, please provide a description of the content that was created and the name of the language model or tool, version and extension numbers, and manufacturer in the space below. (Note: this does not include basic tools for checking grammar, spelling, references, etc.)

5. PNAS journals, 21 February 2023

5.1 For authors

<https://www.pnas.org/post/update/pnas-policy-for-chatgpt-generative-ai>

<https://www.pnas.org/author-center/editorial-and-journal-policies#authorship-and-contributions>

<https://academic.oup.com/pnasnexus/pages/general-instructions>

Use of artificial intelligence (AI) software, such as ChatGPT, must be noted in the Materials and Methods section (or Acknowledgments, if no Materials and Methods section is available) of the manuscript and may not be listed as an author. Authors are solely accountable for, and must thoroughly fact-check, outputs created with the help of generative AI software.

5.2 For reviewers

<https://www.pnas.org/reviewer>

<https://academic.oup.com/pnasnexus/pages/for-reviewers>

Reviewers may not submit any manuscript or related materials, including their own comments about the work, to any artificial intelligence (AI) software, such as ChatGPT.

6. Nature and Springer Nature Journals, 24 January 2023

6.1 For authors

<https://www.nature.com/articles/d41586-023-00191-1>

<https://www.nature.com/nature/for-authors/initial-submission>

<https://www.nature.com/nature-portfolio/editorial-policies/ai>

Large Language Models (LLMs), such as ChatGPT, do not currently satisfy our authorship criteria. Notably an attribution of authorship carries with it accountability for the work, which cannot be effectively applied to LLMs. Use of an LLM should be properly documented in the Methods section (and if a Methods section is not available, in a suitable alternative part) of the manuscript.

Generative AI Images

While legal issues relating to AI-generated images and videos remain broadly unresolved, Springer Nature journals are unable to permit its use for publication.

Exceptions are images/art obtained from agencies that we have contractual relationships with that have created images in a legally acceptable manner. Other exceptions to this policy include images and video that are directly referenced in a piece that is specifically about AI and will be reviewed on a case-by-case basis.

As we expect things to develop rapidly in this field in the near future, we will review this policy regularly and adapt it if necessary.

Please note: Not all AI tools are generative. The use of non-generative machine learning tools to manipulate, combine or enhance existing images or figures should be disclosed in the relevant caption upon submission to allow a case-by-case review.

6.2 For reviewers

<https://www.nature.com/nature-portfolio/editorial-policies/confidentiality>

<https://www.nature.com/nature/for-referees/policies-and-processes>

Not available.

7. The Lancet group, June 2023

7.1 For authors

<https://www.thelancet.com/preparing-your-manuscript>

The use of AI and AI-assisted technologies in scientific writing

Where authors use AI and AI-assisted technologies in the writing process, these technologies should only be used to improve readability and language of the work and not used to replace researcher tasks such as producing scientific insights, analysing and interpreting data, or drawing scientific conclusions. Applying these technologies should only be done with human oversight and control, and authors should carefully review and edit the result because AI can generate authoritative-sounding output that can be incorrect, incomplete, or biased. Authors should not list AI and AI-assisted technologies as an author or co-author, nor cite AI as an author. Authors are ultimately responsible and accountable for the originality, accuracy, and integrity of the work; and should disclose the use of AI and AI-assisted technologies in a statement at the end of the article.

7.1 For reviewers

<https://www.thelancet.com/peer-review>

Reviewers should not use Artificial Intelligence (AI) or AI-assisted technologies to assist in the scientific review of a paper as the critical thinking and original assessment needed for peer review is outside of the scope of this technology, and there is a risk that the technology will generate incorrect, incomplete or biased conclusions about the manuscript.

Peer reviewers must treat the papers that editors share with them as confidential. Reviewers should not upload the paper, or any part of it, into an AI tool as this may violate the authors' confidentiality and proprietary rights and, where the paper contains personally identifiable information, may breach data privacy rights. This confidentiality requirement extends to peer reviewers' comments, because they may contain confidential information about the paper and/or the authors. For this reason, reviewers should not upload their peer review comments into an AI tool, even if it is just for the purpose of improving language and readability.

Peer reviewers are ultimately responsible and accountable for the originality, accuracy, and integrity of the reviewer comments.

8. Science family of journals, 16 November 2023 (updated from 26 January 2023)

8.1 For authors

<https://www.science.org/content/page/science-journals-editorial-policies>

<https://www.science.org/content/blog-post/change-policy-use-generative-ai-and-large-language-models>

AI-assisted technologies [such as large language models (LLMs), chatbots, and image creators] do not meet the Science journals' criteria for authorship and therefore may not be listed as authors or co-authors, nor may sources cited in Science journal content be authored or coauthored by AI tools. Authors who use AI-assisted technologies as components of their research study or as aids in the writing or presentation of the manuscript should note this in the cover letter and in the acknowledgments section of the manuscript. Detailed information should be provided in the methods section: The full prompt used in the production of the work, as well as the AI tool and its version, should be disclosed. Authors are accountable for the accuracy of the work and for ensuring that there is no plagiarism. They must also ensure that all sources are appropriately cited and should carefully review the work to guard against bias that may be introduced by AI. Editors may decline to move forward with manuscripts if AI is used inappropriately. Reviewers may not use AI technology in generating or writing their reviews because this could breach the confidentiality of the manuscript.

AI-generated images and other multimedia are not permitted in the Science journals without explicit permission from the editors. Exceptions may be granted in certain situations—e.g., for images and/or videos in manuscripts specifically about AI and/or machine learning. Such exceptions will be evaluated on a case-by-case basis and should be disclosed at the time of submission. The Science journals recognize that this area is rapidly developing, and our position on AI-generated multimedia may change with the evolution of copyright law and industry standards on ethical use.

<https://www.science.org/doi/10.1126/science.adg7879>

For years, authors at the *Science* family of journals have signed a license certifying that “the Work is an *original*” (italics added). For the *Science* journals, the word “original” is enough to signal that text written by ChatGPT is not acceptable: It is, after all, plagiarized from ChatGPT. Further, our authors certify that they themselves are accountable for the research in the paper. Still, to make matters explicit, we are now updating our license and Editorial Policies to specify that text generated by ChatGPT (or any other AI tools) cannot be used in the work, nor can figures, images, or graphics be the products of such tools. And an AI program cannot be an author. A violation of these policies will constitute scientific misconduct no different from altered images or plagiarism of existing works. Of course, there are many legitimate data sets (not the text of a paper) that are intentionally generated by AI in research papers, and these are not covered by this change.

Artificial intelligence (AI). Text generated from AI, machine learning, or similar algorithmic tools cannot be used in papers published in Science journals, nor can the accompanying figures, images, or graphics be the products of such tools, without explicit permission from the editors. In addition, an AI program cannot be an author of a Science journal paper. A violation of this policy constitutes scientific misconduct.

8.1 For reviewers

<https://www.science.org/content/page/science-information-reviewers>

Artificial intelligence (AI). Use of large language models and other generative AI tools is not allowed. The reviewer is responsible for writing their own review and should not solicit input from colleagues without first obtaining permission from the editor.

Templates of AI policies for authors and reviewers

An AI policy for authors

Authorship and responsibility: AI systems are ineligible for authorship. Authors must assume full responsibility for the submitted work and ensure ethical and legal compliance when using AI. Any AI-generated content should be carefully reviewed for accuracy, plagiarism, and bias, and properly attributed.

Reporting: Should AI tools be employed beyond the scope typically covered by university writing centers or professional editing services, authors must disclose such use.

What to include:

- Name, version, and developer of the AI tool(s) used (e.g., ChatGPT, version as of September 25, based on GPT-4, developed by OpenAI).

How to report:

- Specify the sections and extent of AI tool use (e.g., “In the Discussion section, approximately 20% of the text was initially drafted by AI.”).
- Outline the type and purpose of generated content that was incorporated (e.g., “The AI-generated text aimed to provide a structured summary of the key findings and their implications. This generated content was later edited and refined by the authors to ensure coherence, accuracy, and relevance.”).
- Provide the prompts given to the AI systems, along with date/time (e.g., a link or a screenshot of the chat).

Where to report:

- For text generation, disclose in the Acknowledgements section.
- For data analysis, code, or results, report in the Methods section.
- If the above is not available, disclose in an appropriate section of the manuscript.

An AI policy for reviewers

Confidentiality: Reviewers must not input manuscript text, data, figures, or their own comments into AI systems unless confidentiality is assured. Note that interactions with ChatGPT, for instance, become part of its training data and thereby breach confidentiality, unless users opt out through “Chat history & training” or employ ChatGPT Enterprise.

Reporting: When using AI beyond what is commonly provided by university writing centers or professional editing services, reviewers should report 1) the name, version, and developer of the AI tool(s) and 2) the specific nature and purpose of the AI usage.

(Note: The policy templates will be translated into multiple languages once the manuscript is finalized.)